OctoAssist

Hosting Infrastructure Requirements

Specification for deploying or migrating the OctoAssist ITSM platform

Third Octopus

Modern Cloud · Trusted Security · Managed Outcomes

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1. Document Control

Field	Detail
Document	OctoAssist — Hosting Infrastructure Requirements
Version	1.0
Date	May 2026
Owner	Third Octopus — Product Engineering
Classification	Private & Confidential
Purpose	Define the infrastructure required to host or migrate OctoAssist to a new environment
Current environment	Co-hosted on the hrms-erp droplet (DigitalOcean BLR1)

2. Current Environment (As-Is)

OctoAssist currently co-hosts on the hrms-erp droplet, sharing the host, the nginx reverse proxy and the certbot TLS automation with the HRMS application.

Layer	Component	Detail
Host	DigitalOcean droplet	Region BLR1 · Ubuntu 24.04.4 LTS · kernel 6.8 · x86-64
Capacity	Shared with HRMS	2 vCPU · 3.8 GB RAM · 77 GB disk (shared, not dedicated)
Runtime	Docker + Compose	Docker 29.2 · Docker Compose v5
App container	octoassist-app	FastAPI on uvicorn · image 353 MB · port 8088 → 8080
DB container	octoassist-postgres	postgres:16-alpine · image 395 MB
Proxy / TLS	hrms-nginx	Owned by HRMS stack · terminates 80/443 · proxies the subdomain
Certificate	Let's Encrypt	certbot on host · auto-renew every 90 days
Public URL	DNS A record	octoassist.thirdoctopus.com

3. Measured Resource Consumption

OctoAssist's own footprint is small. Figures below are steady-state at idle for the current load (43 endpoints, 399 users).

Resource	App	Postgres	Total
RAM	119 MB	66 MB	**≈ 185 MB**
CPU (of 2 vCPU)	0.7 %	0.1 %	**< 1 %**
Database size	—	18 MB	**18 MB**
Disk (images + DB + uploads)	353 MB	395 MB	**< 1.5 GB**

Note: the two background threads (SLA escalator every 5 min, patch scheduler every 60 s) are the only steady-state work. The database grows slowly — well under 1 GB even at 10× the current endpoint count.

4. Provisioning Specification (CPU · RAM · Disk · Bandwidth)

Headline specification when OctoAssist runs on its own server rather than co-hosted. Three tiers; Recommended is the default choice for the current and near-term load.

Specification	Minimum	Recommended	Comfortable
CPU	1 vCPU (x86-64)	**2 vCPU (x86-64)**	2–4 vCPU (x86-64)
RAM	2 GB	**4 GB**	8 GB
HDD type	SSD (required)	**NVMe SSD (preferred)**	NVMe SSD
HDD size	20 GB	**40 GB**	80 GB
Bandwidth (link)	10 Mbps	**25 Mbps**	50–100 Mbps
Monthly transfer	≈ 50 GB	**≈ 150 GB**	250 GB+
Suitable for	Current load (43 endpoints)	**Up to ~500 endpoints**	1,000+ endpoints

Constraint	Requirement
CPU architecture	x86-64 (amd64) — the image is not multi-arch; an ARM host needs the image rebuilt for arm64 first
Disk type	SSD is required, not spinning HDD — Postgres + container I/O need low-latency random reads. NVMe preferred.
Operating system	Any Docker-capable Linux; Ubuntu 22.04 / 24.04 LTS recommended

5. Bandwidth — Detail

Public bandwidth demand is low. The figures below are conservative budgets for the current 43-endpoint scale; scale roughly linearly with endpoint count.

Traffic source	Direction	Volume	Notes
Agent check-ins	Inbound	Low / steady	43 endpoints poll every 30 s — small JSON (1–2 KB each) + periodic asset snapshot (10–50 KB)
Web UI (admin/agent)	Both	Low / bursty	Pages 20–450 KB; only a handful of concurrent staff users
Email (Graph)	Outbound	Very low	5–20 KB per notification, sent to graph.microsoft.com
TLS + ACME renewal	Both	Negligible	Cert renewal every 90 days

Traffic source	Direction	Volume	Notes
Software / patch payloads	Outbound	Bursty (only if self-hosted)	Installers are normally fetched by endpoints DIRECT from vendor URLs — they do NOT transit OctoAssist unless you use the built-in upload-and-host pipeline

> **Sizing rule:** sustained public throughput stays under 1 Mbps average with 5–10 Mbps peaks. A 10 Mbps link is sufficient; 25 Mbps gives comfortable headroom. Add the installer size × endpoint count to the monthly transfer budget only if you host installers on OctoAssist rather than linking to vendor URLs.

6. Baseline Operating System

The application runs entirely inside a container, so the host OS only needs to run Docker, nginx and certbot. The current production baseline is below; alternatives are listed for portability.

Item	Baseline / Requirement
Operating system	Ubuntu 24.04 LTS (Noble Numbat) — current production baseline. Ubuntu 22.04 LTS also fully supported.
Kernel	Linux 6.x (production host runs 6.8.0-71-generic)
Init system	systemd (used to run Docker + the nginx/certbot services at boot)
CPU architecture	x86-64 (amd64) — mandatory unless the image is rebuilt for arm64
Filesystem	ext4 or xfs on SSD/NVMe; Docker overlay2 storage driver (default)
Supported alternatives	Any modern Docker-capable Linux — Debian 12, RHEL 9, Rocky / AlmaLinux 9, Amazon Linux 2023. The container itself is OS-agnostic.
Not recommended	ARM/arm64 hosts (need image rebuild); Windows Server via Docker Desktop / WSL2 (not production-grade for this workload)

7. Software Prerequisites (Host)

Everything Python-related lives inside the container; the host needs only the items below.

Component	Minimum version	Role
Docker Engine	24+ (host runs 29.2)	Container runtime
Docker Compose	v2+ (host runs v5)	Orchestrates the app + Postgres stack
nginx	Any current	Reverse proxy + TLS termination (or a managed load balancer)
certbot	Any current	Let's Encrypt certificate issue + auto-renew
git	Any current	Clone the repository to build the image

8. Access & Administrative Requirements

Deploying and operating OctoAssist needs access at three levels: the host, the DNS / network, and the Microsoft 365 tenant (for SSO and mail). Each is listed below with who needs it and why.

Access	Level	Held by	Purpose
SSH — root or sudo	Host	Platform / IT admin	Install Docker, run compose, certbot, OS patching, log review
SSH on port 22	Host	Restricted to admin IPs	Key-based auth only; must NOT be open to the public internet
Docker group membership	Host	Deploy user (optional)	Run docker / compose without full root — hardening option
DNS record management	Network	Owner of thirdoctopus.com DNS	Create or repoint the A record for the hostname
Firewall / security group	Network	Network admin	Open inbound 80 + 443; allow outbound to Microsoft + Let's Encrypt
Entra ID directory role	M365	Global Admin / App Admin	Manage the SSO app registration, grant Mail.Send admin consent, rotate the client secret
Licensed mailbox	M365	helpdesk@temaindia.com	Send-as identity for Microsoft Graph mail
OctoAssist bootstrap admin	App	ADMIN_USERNAME / PASSWORD	First login; create and assign all other users / roles
Git / registry read	Build	Deploy user	Clone the repository (or pull the image) to build on the host

> **Note:** the host-level (SSH) access and the Microsoft 365 (Entra) access are usually held by different teams. Plan for both before the migration window — the SSO + mail integration cannot be completed without an Entra administrator on hand.

9. Network, DNS, TLS & Ports

Item	Requirement
Public DNS	A record for the hostname → new host public IP
Inbound 443/tcp	HTTPS — primary application traffic (required)
Inbound 80/tcp	HTTP → HTTPS redirect + ACME (Let's Encrypt) challenge
Other inbound	None — nothing else needs to be public
TLS certificate	Let's Encrypt (free, auto-renew) or own cert — HTTPS is mandatory; agents reject plain HTTP
Outbound egress	login.microsoftonline.com, graph.microsoft.com (SSO + mail); letsencrypt.org (certs); GitHub / PyPI (image build)

> **Critical:** Keep the hostname octoassist.thirdoctopus.com. All 43 endpoint agents and the Entra SSO redirect URI are bound to it — repoint the DNS A record and the migration is transparent. Change the hostname and every agent must be re-pointed and SSO reconfigured.

10. Persistent Data to Migrate

Exactly three artifacts carry state. Lose any one and data or access is lost.

#	Artifact	Location	Why it is critical
1	Postgres data	Docker volume octoassist_postgres_data	All tickets, changes, assets, users, SLAs, KB, audit trail
2	Uploads	Host dir /opt/octoassist/uploads	Ticket attachments
3	Encryption key	OCTOASSIST_ENCRYPTION_KEY in .env	Fernet key that decrypts the Graph mail secret + SSO secrets in the DB

Migration in one line: pg_dump the database, copy the uploads directory, and carry the encryption key to the new host.

11. Environment Variables & Secrets

Read from deploy/docker/.env. Graph mail and SSO secrets are NOT here — they live encrypted in the database.

Variable	Purpose	Carry over?
POSTGRES_PASSWORD	Postgres password (internal only)	May be regenerated
ADMIN_USERNAME	Bootstrap admin login	Optional
ADMIN_PASSWORD	Bootstrap admin password	Optional
TENANT_NAME	Display name (Tema IT Team)	Yes
OCTOASSIST_ENCRYPTION_KEY	Fernet key for secrets at rest	**MUST carry verbatim**

12. External Dependencies

Service	Used for	Migration impact
Microsoft Entra ID	Single sign-on	No change if hostname unchanged; else update OIDC redirect URI
Microsoft Graph (Mail.Send)	Email notifications	No change — credentials travel inside the DB dump
Let's Encrypt	TLS certificate	Re-issue on new host (automatic via certbot)
Enrolled Windows endpoints	Inventory + actions	No change if hostname unchanged; else re-point all agents

13. Backup & Disaster Recovery

Item	Recommendation
Database backup	Nightly pg_dump to off-box storage (S3 / Spaces), 30-day retention — DB is ~18 MB
Uploads backup	Sync /opt/octoassist/uploads to the same off-box storage
Encryption key	Store separately in a password manager / secrets vault — never alongside the DB backup
Recovery time	Full rebuild on a fresh host ≈ 15 minutes (clone repo → compose up → restore dump)

14. Summary

Question	Answer
Smallest viable host	1 vCPU · 2 GB RAM · 20 GB SSD · 10 Mbps · x86-64 Linux
Recommended host	2 vCPU · 4 GB RAM · 40 GB NVMe SSD · 25 Mbps
Disk type	SSD required (NVMe preferred) — never spinning HDD
Bandwidth	10 Mbps min / 25 Mbps recommended; ≈ 50–150 GB transfer per month
Host software	Docker + Docker Compose + nginx + certbot + git
Public ports	80 and 443 only
What to migrate	Postgres dump + uploads folder + encryption key
Transparent cutover	Keep hostname octoassist.thirdoctopus.com and repoint DNS
Endpoint impact	Zero if hostname unchanged (43 agents keep working)